F005 v0.5.0 - E0 provenance audit

DEVELOPMENT / SOURCE EVIDENCE ONLY / NO EULER DATA VALIDATION

1. Trigger

The user-local v0.5.0 run reports BLOCKED_GEOMETRY_UNVERIFIED. The immediate blocker is missing acceleration_axis_sign and measurement_height_sign.

2. Recovered physical inputs

Camilotto thesis Table 1.1 (printed p.15): internal length 0.080 m, width 0.031 m and filling depth 0.043 m. Table 2.1 (p.34): model density 998 kg/m3 and gravity 9.81 m/s2; capillarity is neglected in the cited model.

3. Why signs cannot simply be guessed

For a linear response with scalar forcing, the predicted exported elevation depends on the product of input and output sign flags. Therefore one experimental input/output series alone cannot distinguish the two physical axis conventions. TRAIN-only comparisons are diagnostics, not independent provenance.

4. Initial condition caution

The thesis describes subtracting the mean of 1000 pre-motion optical samples and a zero-state assumption for the analytical model. Neither proves zero displacement and zero velocity at the exact first sample of every exported 1500-point trajectory.

5. Audit intervention

The supplied script verifies geometry, split and input hashes; inspects only 3400 locked TRAIN response trajectories; compares both relative-polarity hypotheses from a declared zero-state model; summarizes the initial height and finite-difference estimates. It never modifies the main configuration.

6. Advancement gates

A source-defined robot +x convention, exported wall-height sign, and verified per-trace initial state are needed for a source-proven geometry-only forecast. Without these, report E0 as blocked and preserve TEST/OOD sealing.

Operational command

run_e0_provenance_audit.cmd

References

D. Camilotto (2026), Politecnico di Milano MSc thesis, hdl.handle.net/10589/253216.
N. Giulietti et al. (2026), Zenodo dataset v1.0.0, doi:10.5281/zenodo.20039757.